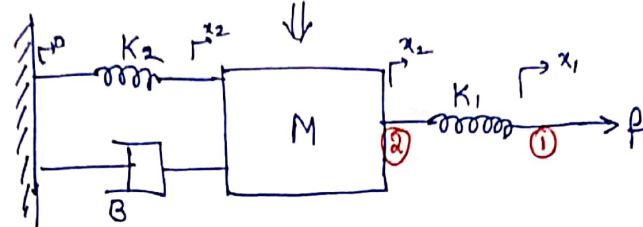
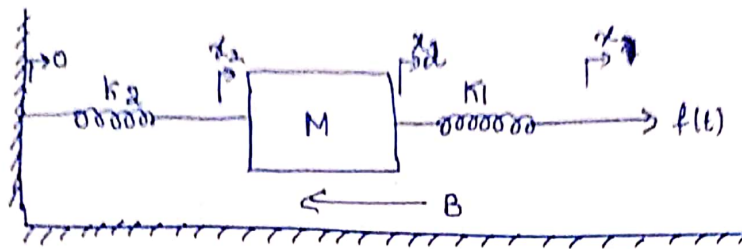


FV- F-I Analogies Problems (Tutorial ①)

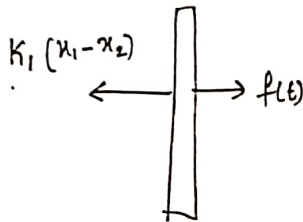
Obtain analogous networks for the following systems based on F.V & F-I analogies (Class 5)

①

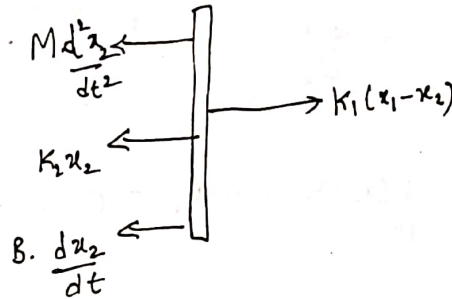
①



FBD at Join ①



FBD at Join ②



$$f = K_1(x_1 - x_2) \rightarrow \textcircled{1}$$

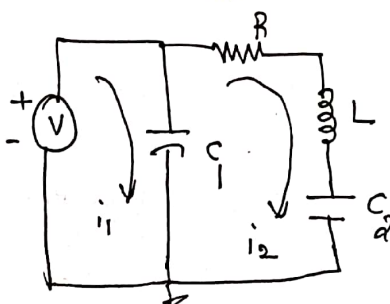
$$M \cdot \frac{d^2x_2}{dt^2} + B \cdot \frac{dx_2}{dt} + K_2x_2 = K_1(x_1 - x_2)$$

$$\text{or } M \cdot \frac{d^2x_2}{dt^2} + B \cdot \frac{dx_2}{dt} + K_2x_2 + K_1(x_2 - x_1) = 0 \rightarrow \textcircled{2}$$

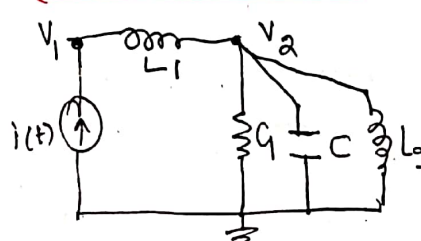
Analogous electrical Eqn for fv-analogy

$$\text{for } \textcircled{1} \quad V = \frac{1}{C_1} \int (i_1 - i_2) \cdot dt \quad (\because K = \frac{1}{C}) \rightarrow \textcircled{1}$$

$$\text{for } \textcircled{2} \quad L \cdot \frac{di_2}{dt^2} + R \cdot i_2 + \frac{1}{C_2} \int i_2 \cdot dt + \frac{1}{C_1} \int (i_2 - i_1) \cdot dt = 0 \rightarrow \textcircled{2}$$



F-I-Analogy (dual n/w)



Duality

$$v(t) \rightarrow i(t)$$

$$R \rightarrow G$$

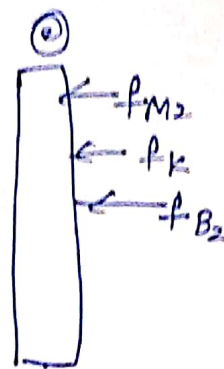
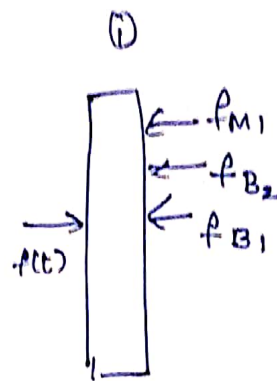
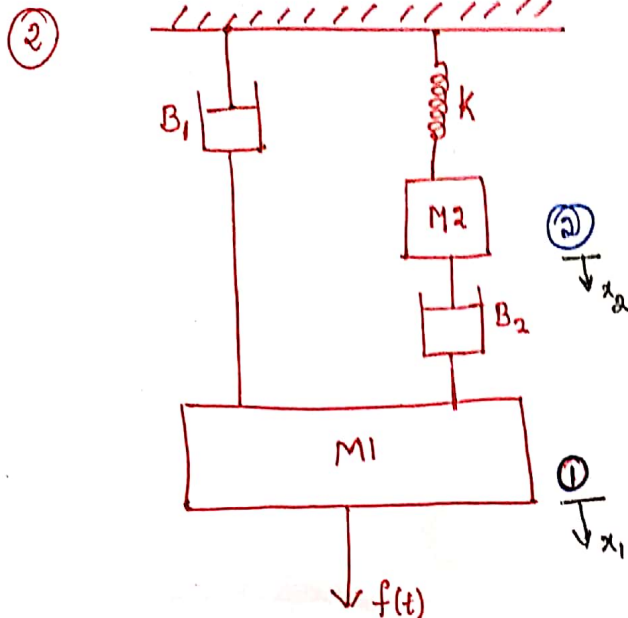
$$C \rightarrow L$$

$$L \rightarrow C$$

$$\text{Series} \rightarrow \text{parallel}$$

$$\text{mesh} \rightarrow \text{nodes}$$

$$\text{mesh} \rightarrow \text{nodes}$$



At Junc ①

$$f(t) = M_1 \frac{d^2 x_1}{dt^2} + B_1 \cdot \frac{dx_1}{dt} + B_2 \frac{d}{dt}(x_1 - x_2) \rightarrow \textcircled{1}$$

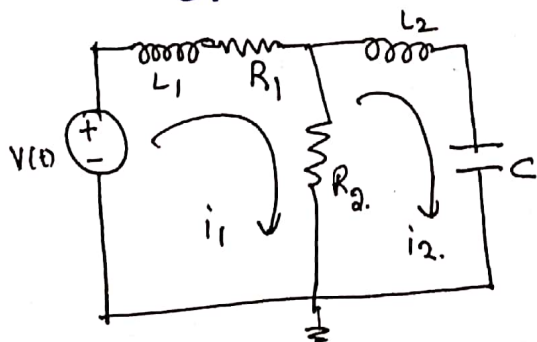
At Junc ②

$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \cdot \frac{d}{dt}(x_2 - x_1) + K \cdot x_2 = 0 \rightarrow \textcircled{2}$$

f-v. analogy.

$$V(t) = L_1 \frac{di_1}{dt} + R_1(i_1) + R_2(i_1 - i_2)$$

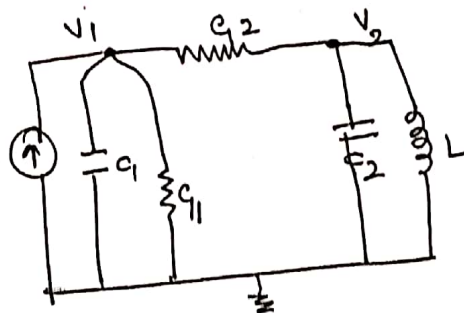
$$0 = L_2 \frac{di_2}{dt} + R_2(i_2 - i_1) + \frac{1}{C} \int i_2 dt$$



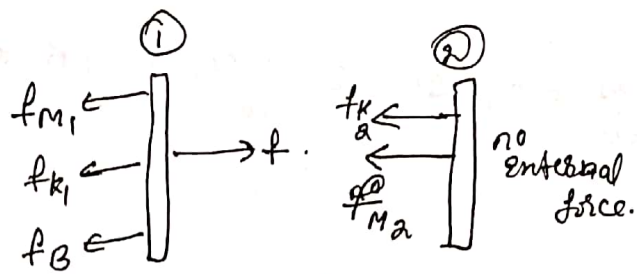
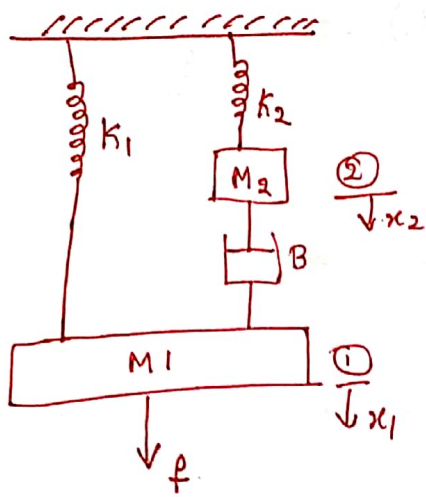
f-I analogy

$$i(t) = \frac{1}{C_1} \int v_1 dt + q_1(v_1) + q_2(v_1 - v_2)$$

$$0 = \frac{1}{C_2} \int v_2 dt + q_2(v_2 - v_1) + L \cdot \frac{d(v_2)}{dt}$$



③



②

At Junc ①

$$f(t) = M_1 \frac{d^2 x_1}{dt^2} + B \frac{d}{dt}(x_1 - x_2) + k_1 x_1 \rightarrow \textcircled{1}$$

At Junc ②

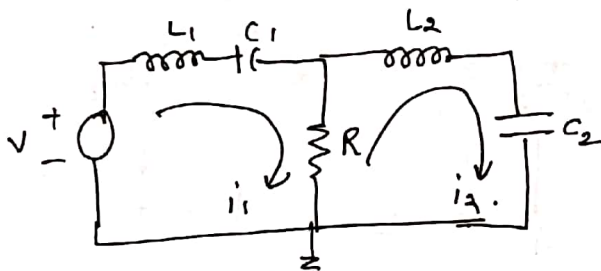
$$B \frac{d}{dt}(x_1 - x_2) = M_2 \frac{d^2 x_2}{dt^2} + k_2 x_2$$

$$\text{or } M_2 \frac{d^2 x_2}{dt^2} + B \frac{d}{dt}(x_2 - x_1) + k_2 x_2 \rightarrow \textcircled{2}$$

FV-Analogy

$$V = L_1 \frac{di_1}{dt} + R(i_1 - i_2) + \frac{1}{C_1} \int i_1 \cdot dt$$

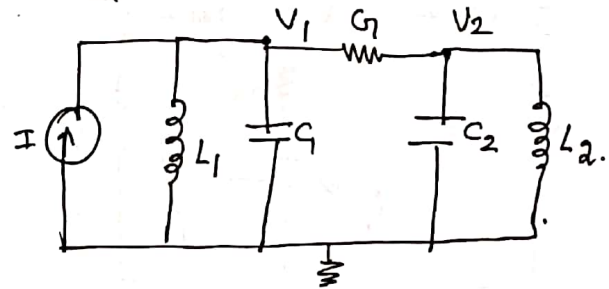
$$0 = L_2 \frac{di_2}{dt} + R(i_2 - i_1) + \frac{1}{C_2} \int i_2 \cdot dt$$



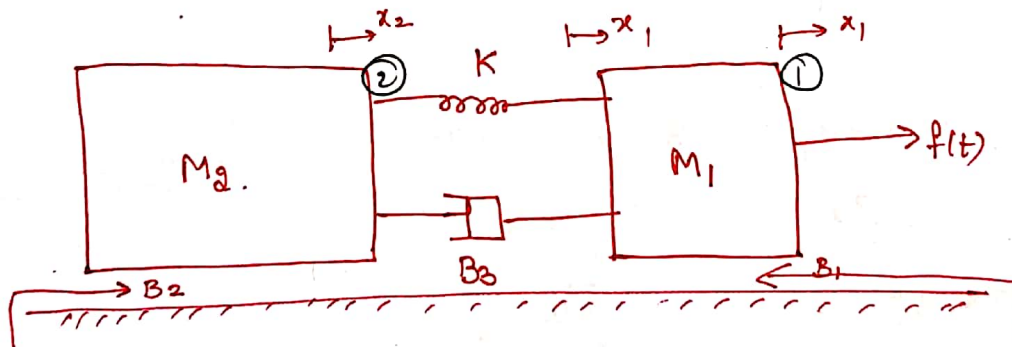
F-I Analogy

$$\mathcal{P} = \frac{1}{C_1} \int V_1 dt + G(V_1 - V_2) + L_1 \frac{dV_1}{dt}$$

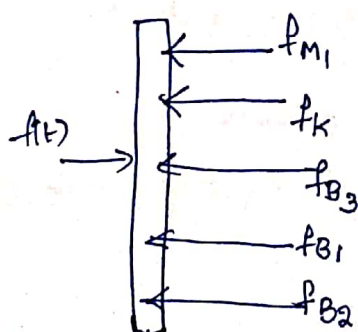
$$0 = \frac{1}{C_2} \int V_2 dt + G(V_2 - V_1) + L_2 \frac{dV_2}{dt}$$



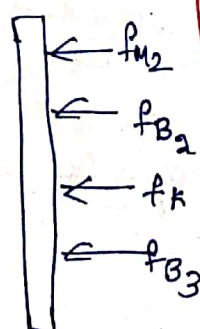
④



Junc ①



Junc ②



① ⇒

$$f(t) = M_1 \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B_3 \frac{d}{dt}(x_1 - x_2) + K(x_1 - x_2) \rightarrow \textcircled{1}$$

②

$$K(x_1 - x_2) + B_3 \frac{d}{dt}(x_1 - x_2) = M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt}$$

or

$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{dx_2}{dt} + B_3 \frac{d(x_2 - x_1)}{dt} + K \cdot (x_2 - x_1) = 0 \rightarrow (2)$$

FV analogy

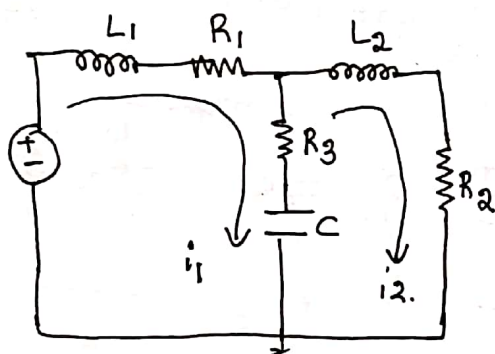
$$V(t) = L_1 \frac{di_1}{dt} + R_1 i_1 + R_3 (i_1 - i_2) + \frac{1}{C} \int (i_1 - i_2) dt \rightarrow (1)$$

$$L_2 \frac{di_2}{dt} + R_2 i_2 + R_3 (i_2 - i_1) + \frac{1}{C} \int (i_2 - i_1) dt = 0 \rightarrow (2)$$

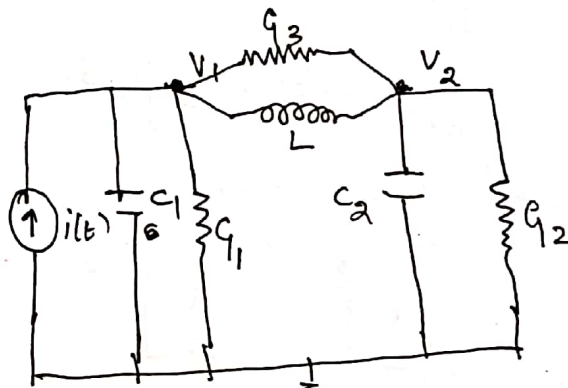
FI analogy

$$i(t) = \frac{1}{C_1} \int V_1 dt + Q_1 (V_1) + Q_2 (V_1 - V_2) + L \frac{d(V_1 - V_2)}{dt}$$

$$\frac{1}{C_2} \int V_2 dt + Q_2 V_2 + Q_3 (V_2 - V_1) + L \frac{d(V_2 - V_1)}{dt} = 0$$

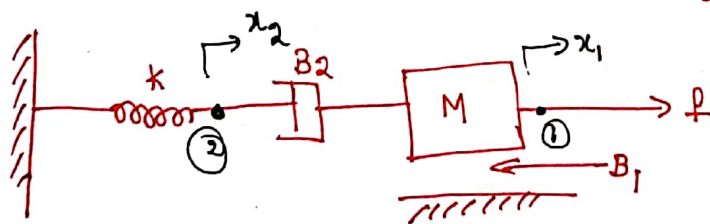


FV analogy

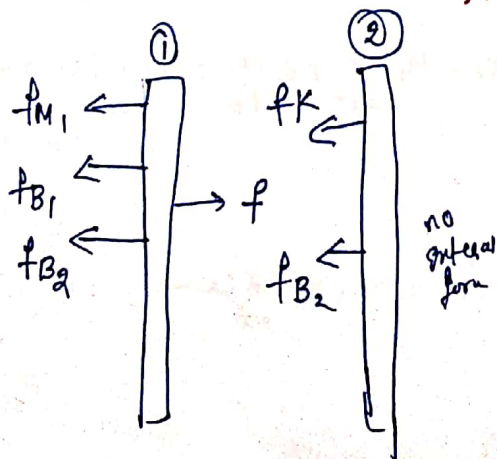


FI analogy

⑤ Draw FV-analogy & F-I analogy



(Note: f is same
 $f = f_1 + f_2$

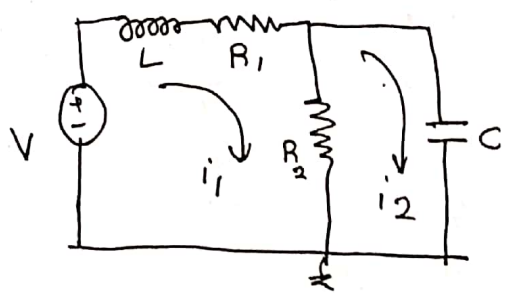


$$f = M \frac{d^2 x_1}{dt^2} + B_1 \frac{dx_1}{dt} + B_2 \frac{d(x_1 - x_2)}{dt} \rightarrow (1)$$

$$B_2 \frac{d}{dt} (x_1 - x_2) = K \cdot x_2 \text{ or}$$

$$B_2 \frac{d}{dt} (x_2 - x_1) + K \cdot x_2 = 0 \rightarrow (2)$$

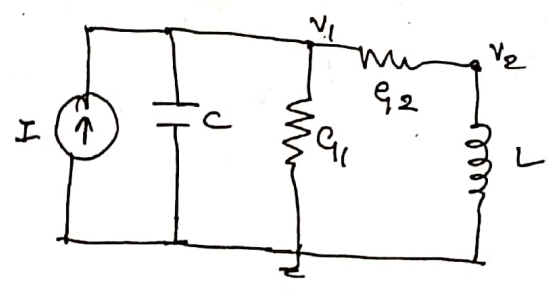
FV Analogy



$$V(t) = L \frac{di_1}{dt} + R_1 i_1 + R_2 (i_1 - i_2) \rightarrow (1)$$

$$R_2 (i_2 - i_1) + \frac{1}{C} \int i_2 \cdot dt = 0 \rightarrow (2)$$

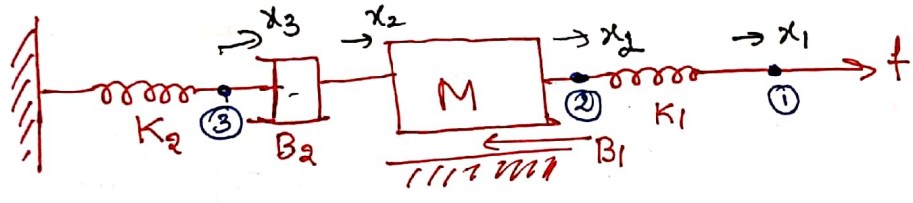
FI Analogy



$$i(t) = \frac{1}{C} \int v_1 \cdot dt + G_1 v_1 + G_2 (v_1 - v_2) \rightarrow (1)$$

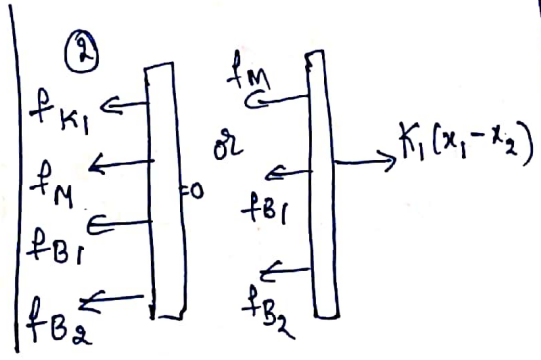
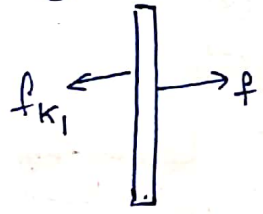
$$G_2 (v_2 - v_1) + L \frac{dv_2}{dt} = 0 \rightarrow (2)$$

6

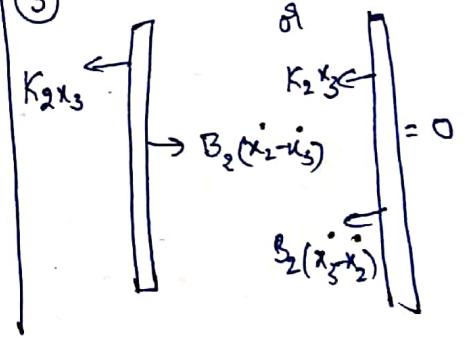


$$f = k_1 (x_1 - x_2)$$

①



③



$$f = k_1 (x_1 - x_2) \rightarrow (1)$$

$$M \cdot \frac{d^2 x_2}{dt^2} + B_1 \frac{dx_2}{dt} + B_2 \frac{d}{dt} (x_2 - x_3) + k_1 (x_2 - x_1) = 0 \rightarrow (2)$$

$$B_2 \frac{d}{dt} (x_3 - x_2) + K_2 x_3 = 0 \rightarrow (3)$$

FV - analogy

$$V(t) = \frac{1}{C_1} \int (i_1 - i_2) dt \rightarrow (1)$$

$$L \frac{di_2}{dt} + R_1 i_2 + R_2 (i_2 - i_3) + \frac{1}{C_1} \int (i_2 - i_1) dt = 0 \rightarrow (2)$$

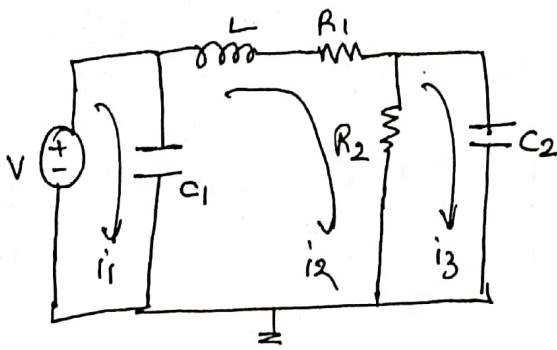
$$R_2 (i_3 - i_2) + \frac{1}{C_2} \int i_3 \cdot dt = 0 \rightarrow (3)$$

F-I - analogy

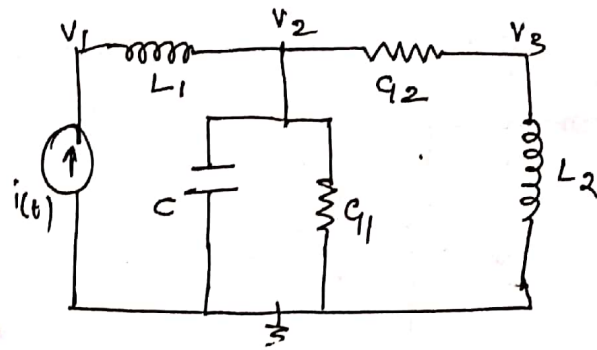
$$i(t) = L_1 \frac{d}{dt} (v_1 - v_2) \rightarrow (1)$$

$$\frac{1}{C} \int v_2 \cdot dt + G_1 v_2 + G_2 (v_2 - v_3) + L_1 \frac{d}{dt} (v_2 - v_1) = 0 \rightarrow (2)$$

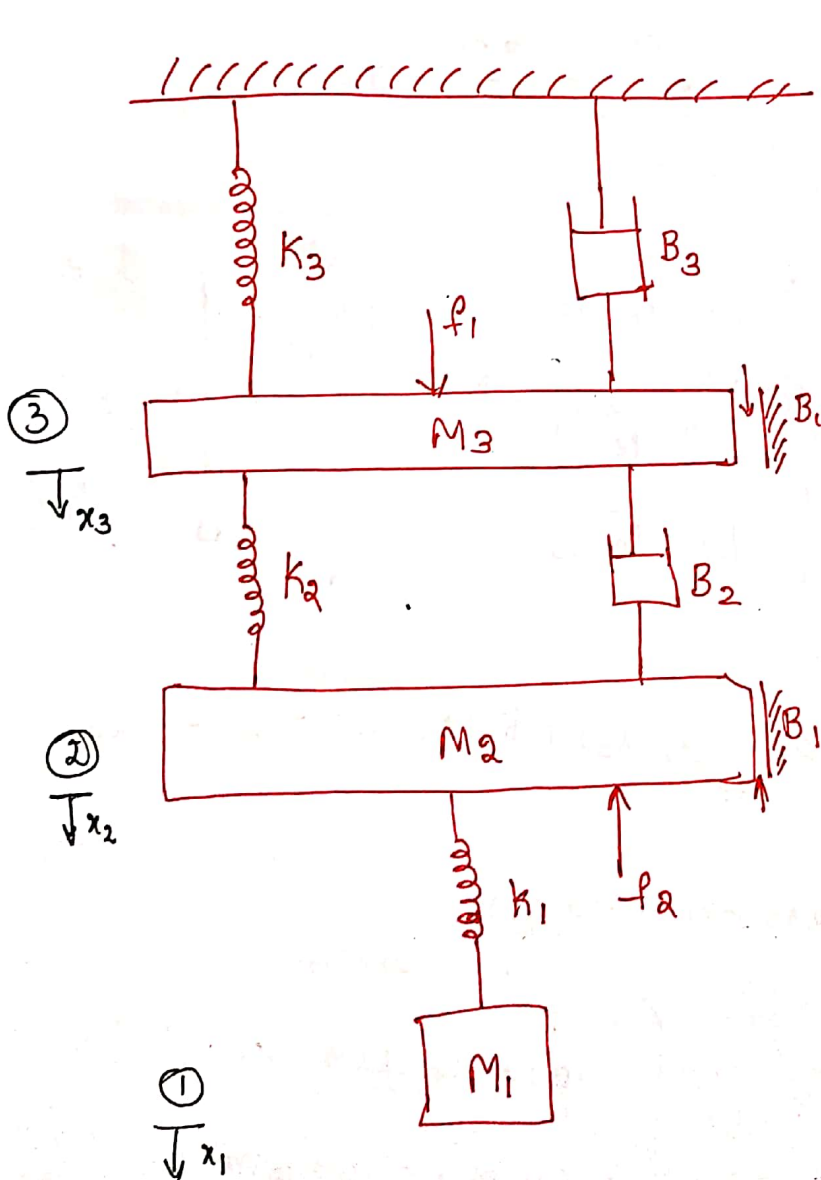
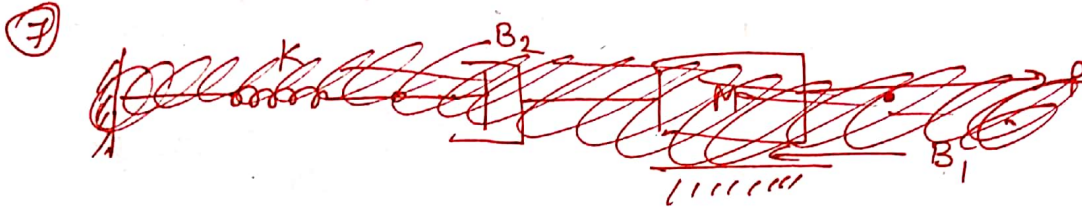
$$G_2 (v_3 - v_2) + L_2 \cdot \frac{dv_3}{dt} = 0$$



FV - Analogy



FI - Analogy



①

$$M_1 \frac{d^2 x_1}{dt^2} + k_1(x_1 - x_2) = 0 \rightarrow \text{①}$$

②

$$M_2 \frac{d^2 x_2}{dt^2} + B_2 \frac{d(x_2 - x_3)}{dt} + B_1 \frac{dx_2}{dt} + k_1(x_2 - x_1) + k_2(x_2 - x_3) = -f_2 \rightarrow \text{②}$$

③

$$M_3 \frac{d^2 x_3}{dt^2} + B_2 \frac{d(x_3 - x_2)}{dt} + B_3 \frac{dx_3}{dt} + k_2(x_3 - x_2) + k_3 x_3 + B_4 \frac{dx_3}{dt} = f_1$$

$$M_3 \frac{d^2 x_3}{dt^2} + B_2 \frac{d}{dt}(x_3 - x_2) + B_3 \frac{dx_3}{dt} + k_2(x_3 - x_2) + k_3 x_3 + B_4 \frac{dx_3}{dt} = f_1$$

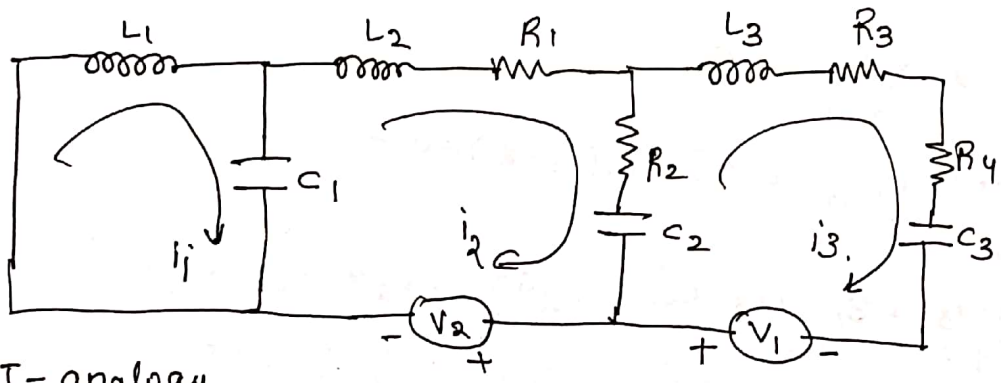
④

FV-analogy

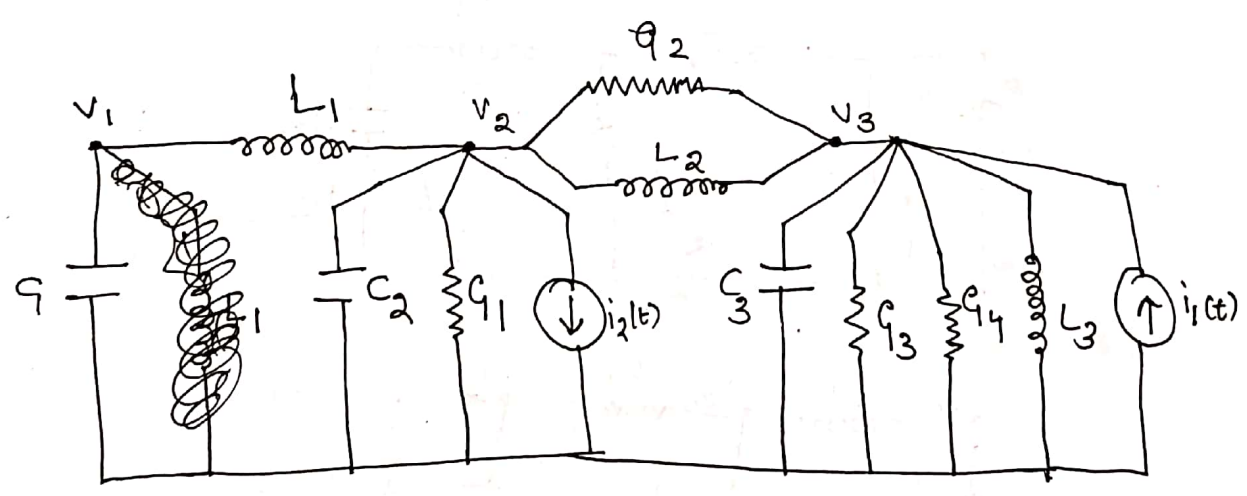
$$L_1 \frac{di_1}{dt} + \frac{1}{C_1} \int (i_1 - i_2) dt = 0 \rightarrow \textcircled{1}$$

$$L_2 \frac{di_2}{dt} + R_2(i_2 - i_3) + R_1 i_2 + \frac{1}{C_1} \int (i_2 - i_1) dt + \frac{1}{C_2} \int (i_2 - i_3) dt = -V_2(t) \rightarrow \textcircled{2}$$

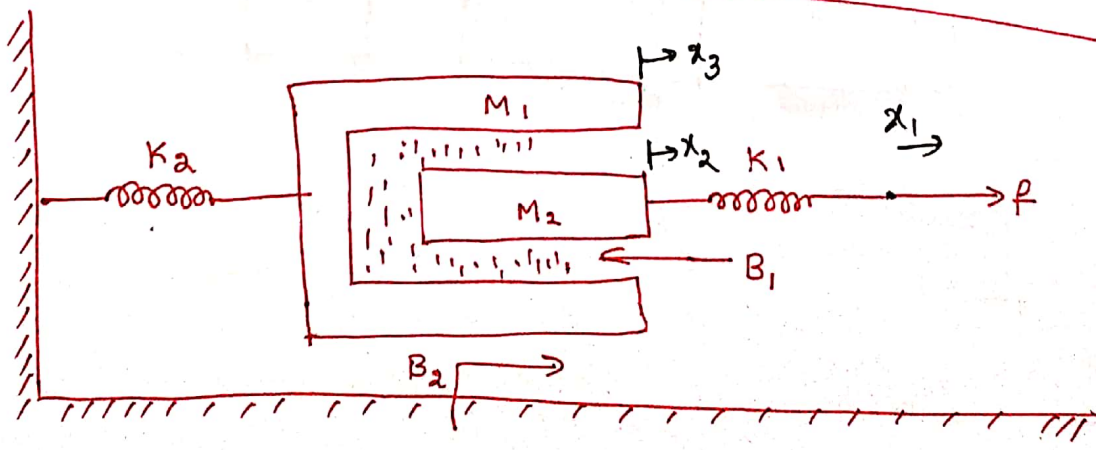
$$V_1(t) = L_3 \frac{di_3}{dt} + R_2(i_3 - i_2) + R_3 i_3 + \frac{1}{C_2} \int (i_3 - i_2) dt + \frac{1}{C_3} \int i_3 dt + R_4 i_3 \rightarrow \textcircled{3}$$

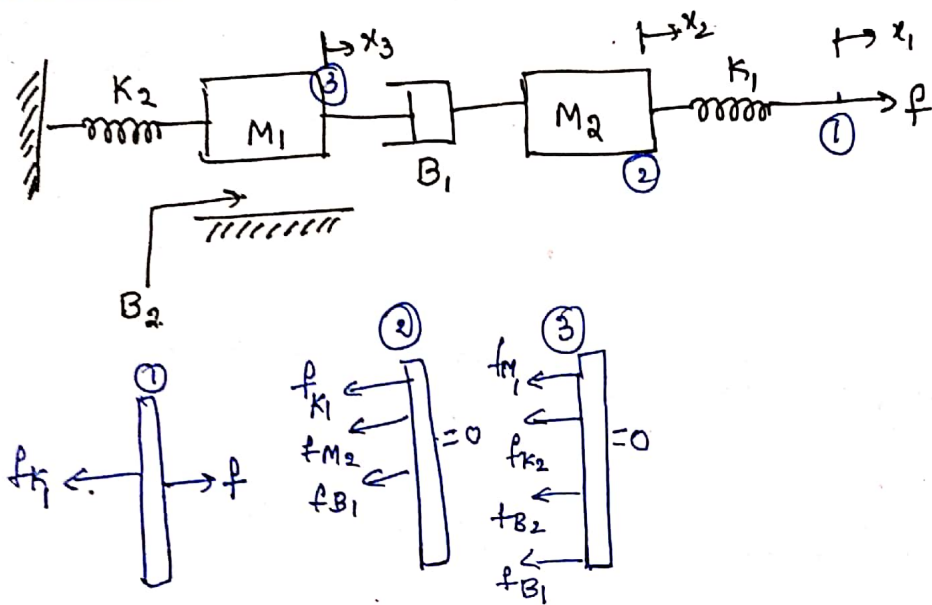


FI-analogy



⑧





$$f = K_1(x_1 - x_2) \rightarrow \textcircled{1}$$

$$v = \frac{1}{c_1} \int (i_1 - i_2) dt$$

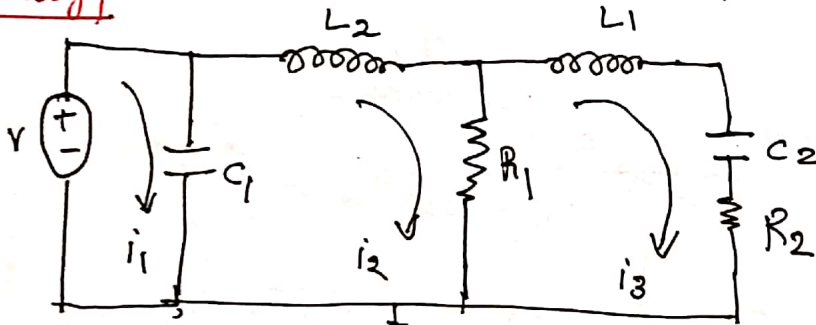
$$M_2 \cdot \frac{d^2 x_2}{dt^2} + B_1 \frac{d}{dt}(x_2 - x_3) + K_1(x_2 - x_1) = 0 \rightarrow \textcircled{2}$$

$$L_2 \frac{di_2}{dt} + R_1(i_2 - i_3) + \frac{1}{c_1} \int (i_2 - i_1) dt = 0$$

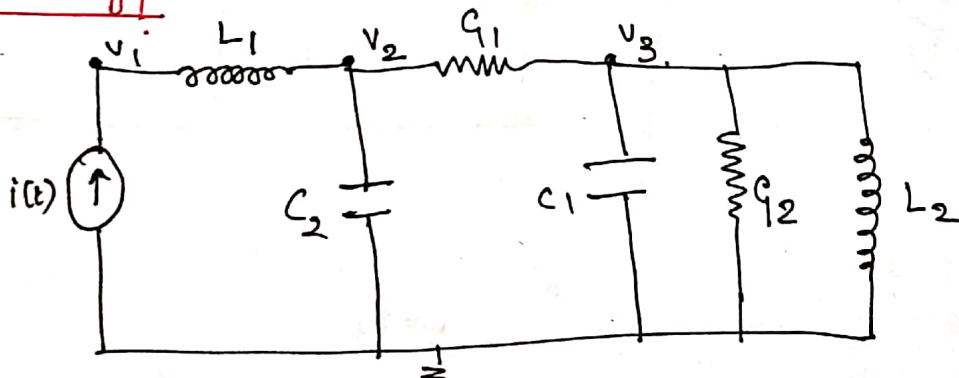
$$M_1 \frac{d^2 x_3}{dt^2} + B_1 \frac{d}{dt}(x_3 - x_2) + B_2 \frac{dx_3}{dt} + K_2 x_3 = 0 \rightarrow \textcircled{3}$$

$$L_1 \frac{di_3}{dt} + R_1(i_3 - i_2) + R_2 i_3 + \frac{1}{c_2} \int i_3 dt = 0$$

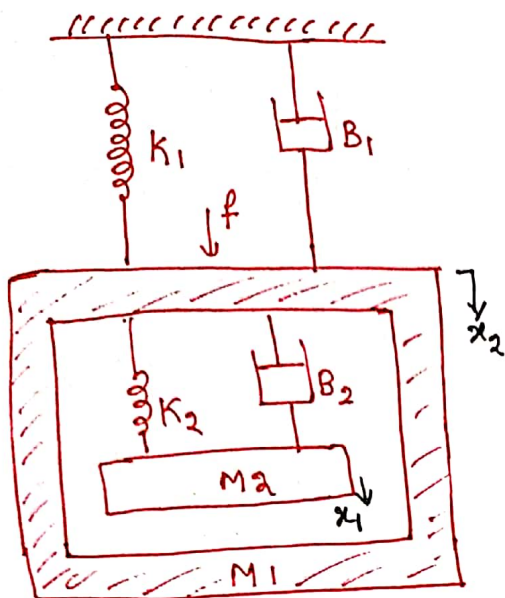
FV-analogy



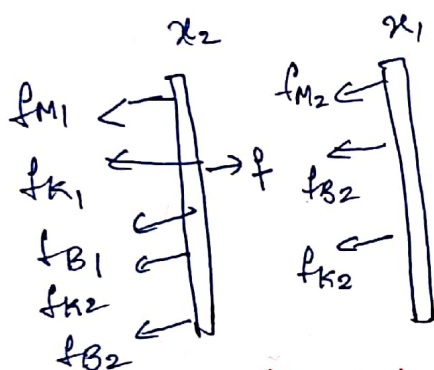
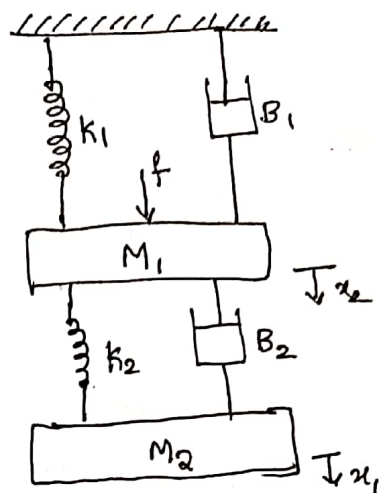
FI-analogy



9



$\Rightarrow$



forces on m_2

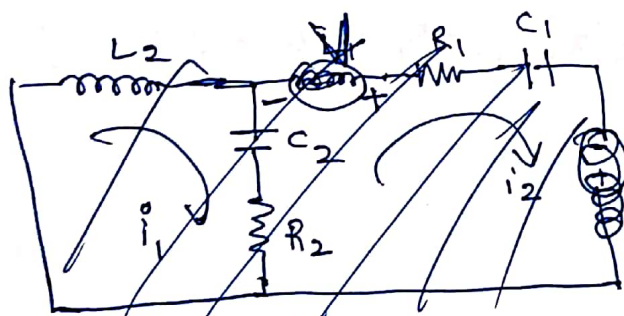
$$L_2 \frac{di_1}{dt} + \frac{1}{C_2} \int (i_1 - i_2) dt + R_2 (i_1 - i_2) = 0$$

$$M_2 \frac{d^2 x_1}{dt^2} + k_2 (x_1 - x_2) + B_2 \frac{d}{dt} (x_1 - x_2) = 0 \rightarrow (1)$$

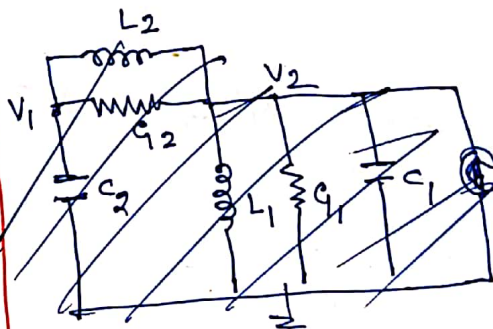
forces on m_1

$$V = L_1 \frac{di_2}{dt} + \frac{1}{C_1} \int i_2 dt + R_1 i_2 + \frac{1}{C_2} \int (i_2 - i_1) dt + R_2 (i_2 - i_1)$$

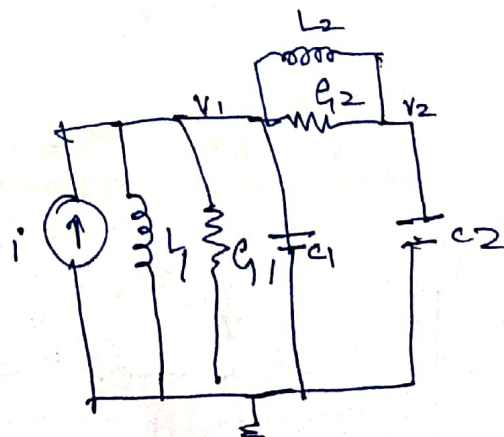
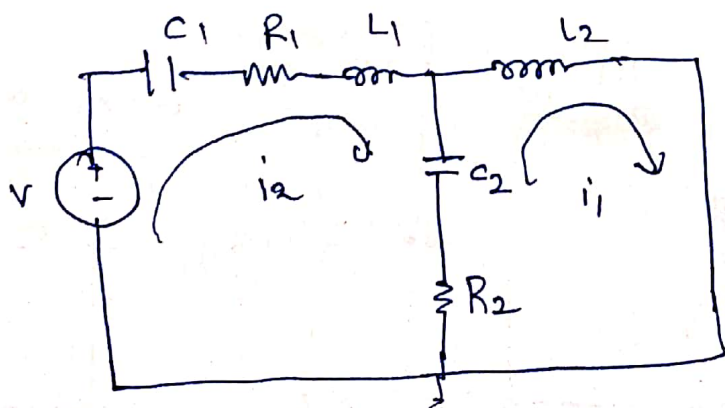
$$f = M_1 \frac{d^2 x_2}{dt^2} + k_1 x_2 + B_1 \frac{dx_2}{dt} + k_2 (x_2 - x_1) + B_2 \frac{d}{dt} (x_2 - x_1) \rightarrow (2)$$



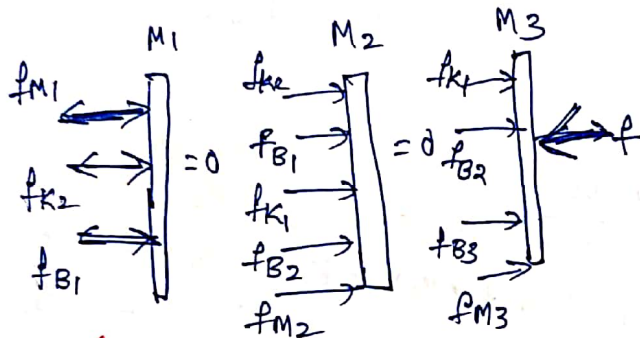
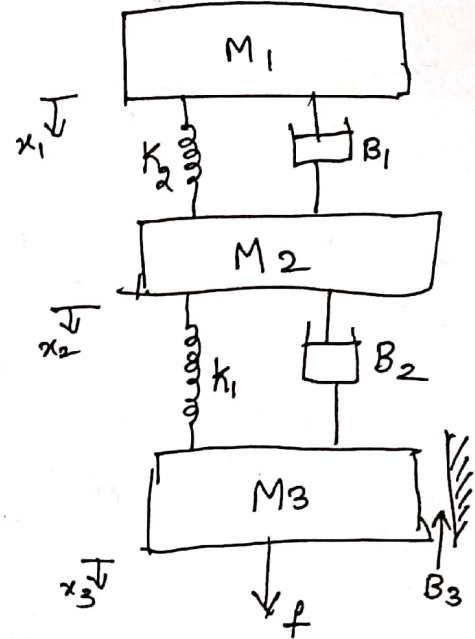
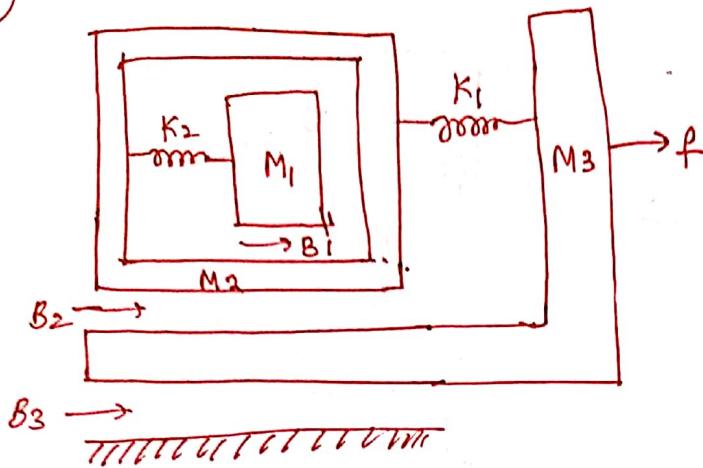
FV analogy



F-I Analogy



10



forces on M1

$$M_1 \frac{d^2 x_3}{dt^2} + K_2(x_3 - x_2) + B_1 \frac{d}{dt}(x_3 - x_2) = 0 \rightarrow (1)$$

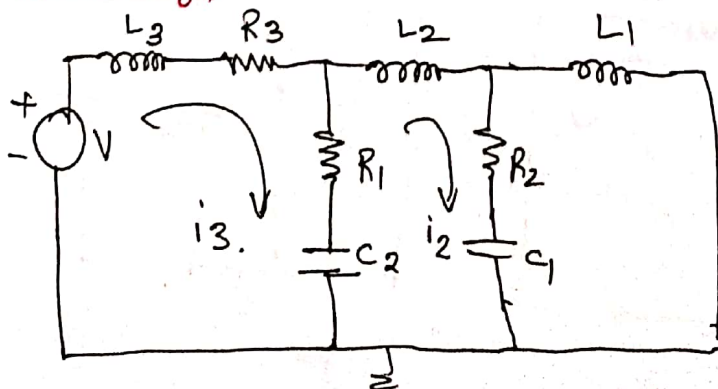
forces on M2

$$M_2 \frac{d^2 x_2}{dt^2} + K_1(x_2 - x_3) + K_2(x_2 - x_1) + B_1 \frac{d}{dt}(x_2 - x_1) + B_2 \frac{d}{dt}(x_2 - x_3) = 0 \rightarrow (2)$$

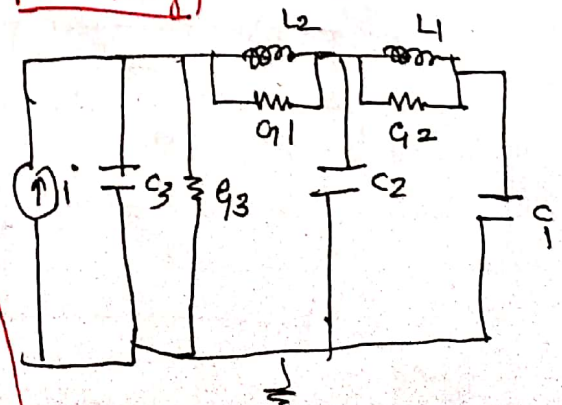
forces on M3

$$M_3 \frac{d^2 x_3}{dt^2} + K_1(x_3 - x_2) + B_2 \frac{d}{dt}(x_3 - x_2) + B_3 \frac{d}{dt}x_3 = f \rightarrow (3)$$

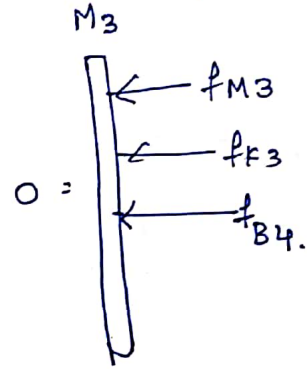
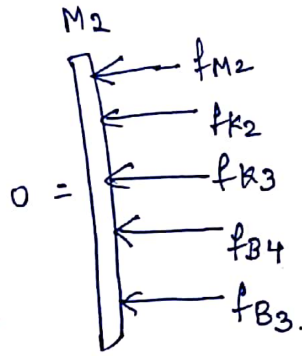
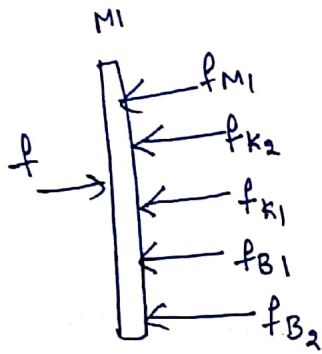
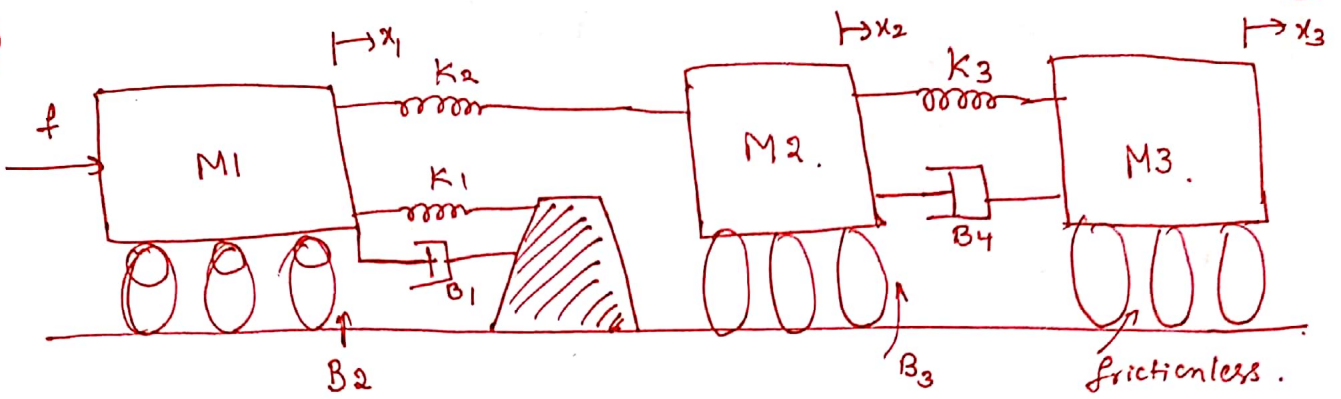
Fv analogy



FI analogy



⑪



$$M_1 \frac{d^2 x_1}{dt^2} + k_2(x_1 - x_2) + k_1 x_1 + B_1 \frac{dx_1}{dt} + B_2 \frac{dx_1}{dt} = f \rightarrow \textcircled{1}$$

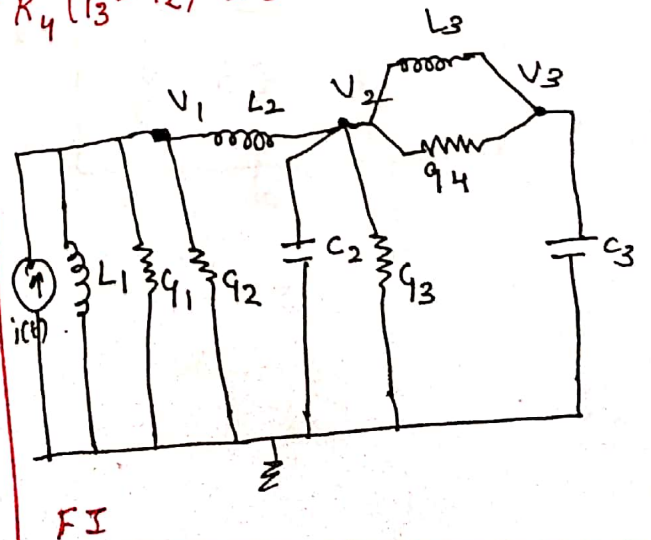
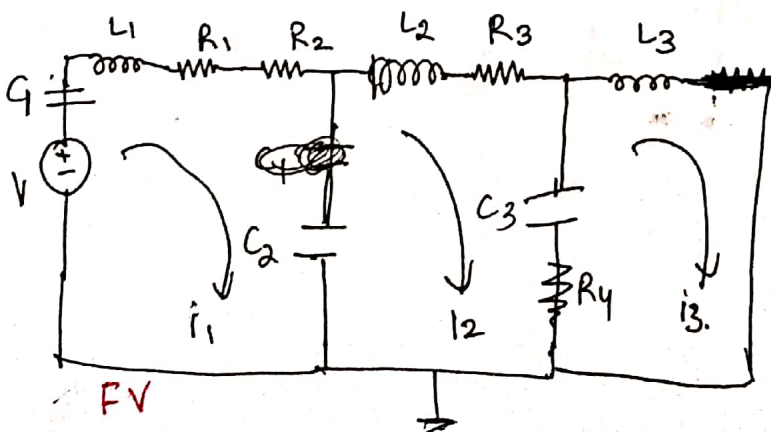
$$L_1 \frac{di_1}{dt} + \frac{1}{C_2} \int (i_1 - i_2) dt + \frac{1}{C_1} \int i_1 dt + R_1 i_1 + R_2 i_1 = V(t)$$

$$M_2 \frac{d^2 x_2}{dt^2} + k_2(x_2 - x_1) + k_3(x_2 - x_3) + B_4 \frac{d(x_2 - x_3)}{dt} + B_3 \frac{dx_2}{dt} = 0$$

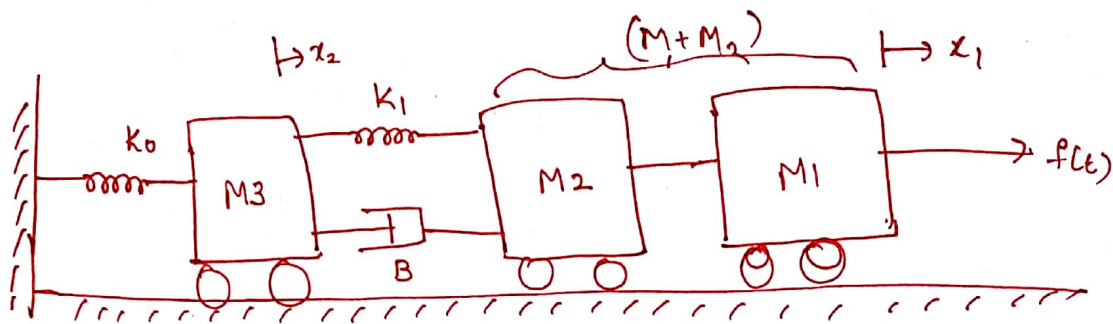
$$L_2 \frac{di_2}{dt} + \frac{1}{C_2} \int (i_2 - i_1) dt + \frac{1}{C_3} \int (i_2 - i_3) dt + R_4(i_2 - i_3) + R_3 i_2 = 0 \rightarrow \textcircled{2}$$

$$M_3 \frac{d^2 x_3}{dt^2} + k_3(x_3 - x_2) + B_4 \frac{d(x_3 - x_2)}{dt} = 0 \rightarrow \textcircled{3}$$

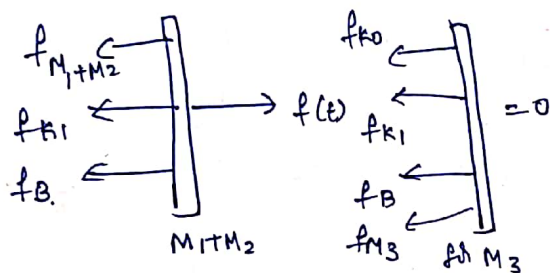
$$L_3 \frac{di_3}{dt} + \frac{1}{C_3} \int (i_3 - i_2) dt + R_4(i_3 - i_2) = 0$$



12



Assume frictionless wheels.



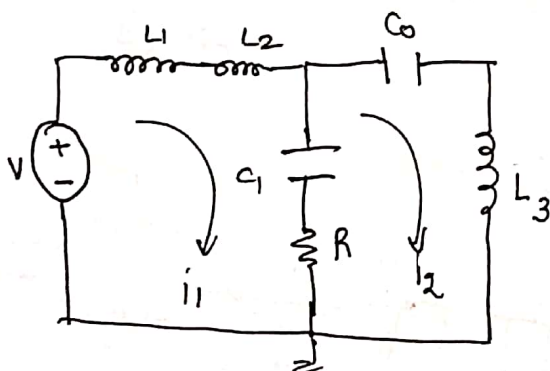
$$M_2 \frac{d^2 x_1}{dt^2} + M_1 \frac{d^2 x_1}{dt^2} + K_1 (x_1 - x_2) + B \frac{d(x_1 - x_2)}{dt} = f(t) \rightarrow (1)$$

$$L_2 \frac{di_1}{dt} + L_1 \frac{di_1}{dt} + \frac{1}{C_1} \int (i_1 - i_2) dt + R (i_1 - i_2) = v(t)$$

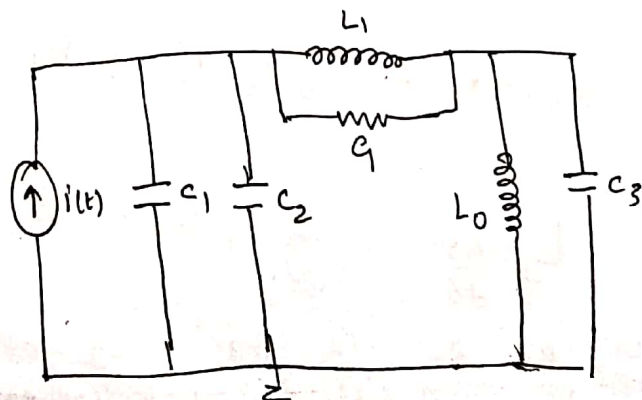
$$K_0 \cdot x_2 + K_1 (x_2 - x_1) + B \frac{d(x_2 - x_1)}{dt} + M_3 \frac{d^2 x_2}{dt^2} = 0 \rightarrow (2)$$

$$\int_0^t i_2 dt + \frac{1}{C_1} \int (i_2 - i_1) dt + R (i_2 - i_1) + L_3 \frac{di_2}{dt} = 0$$

f-v analogy

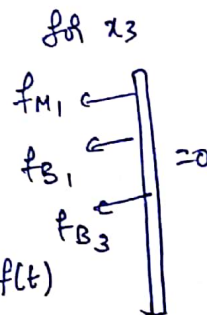
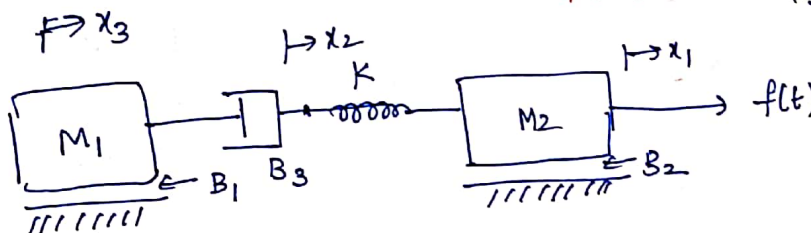
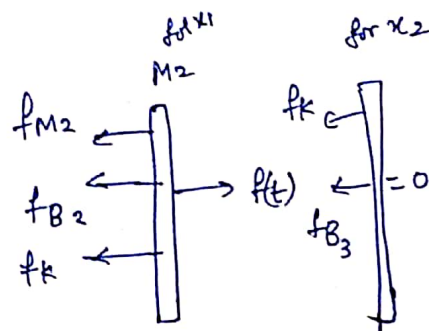
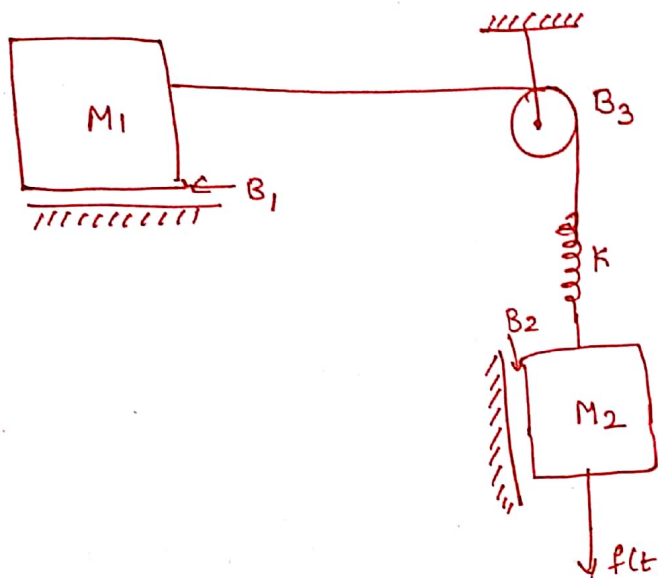


p-I analogy



13.

7



$$M_2 \frac{d^2 x_1}{dt^2} + B_2 \frac{dx_1}{dt} + K(x_1 - x_2) = f(t) \rightarrow (1)$$

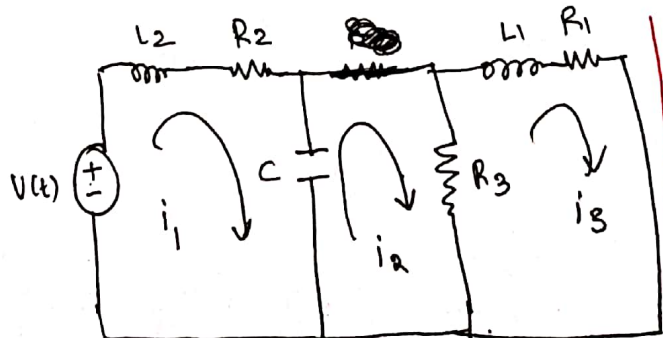
$$L_2 \frac{di_1}{dt} + R_2 i_1 + \frac{1}{C} \int (i_1 - i_2) dt = \phi(t)$$

$$K(x_2 - x_1) + B_3 \frac{d(x_2 - x_3)}{dt} = 0 \rightarrow (2)$$

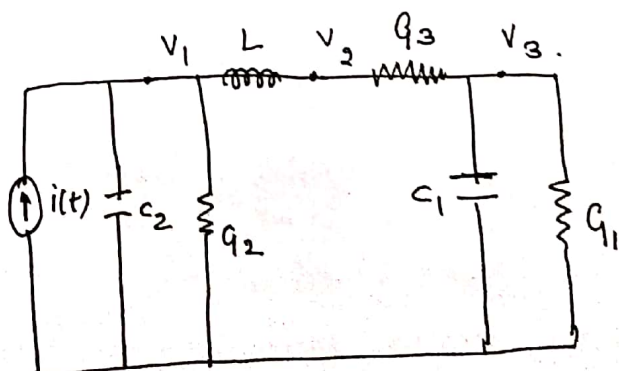
$$\frac{1}{C} \int (i_2 - i_1) dt + R_3 (i_2 - i_3) = 0$$

$$M_1 \frac{d^2 x_3}{dt^2} + B_1 \frac{dx_3}{dt} + B_3 \frac{d(x_3 - x_2)}{dt} = 0 \rightarrow (3)$$

$$L_1 \frac{di_3}{dt} + R_1 i_3 + R_3 (i_3 - i_2) = 0$$

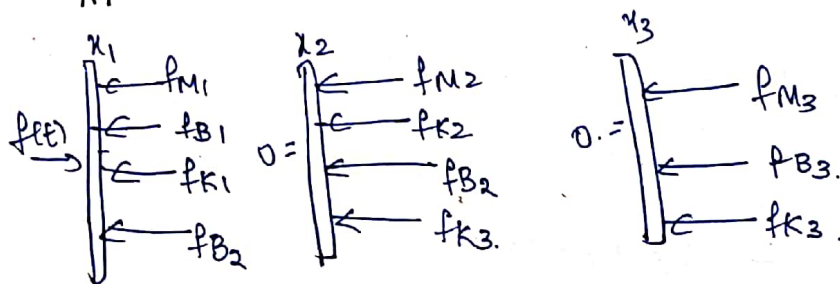
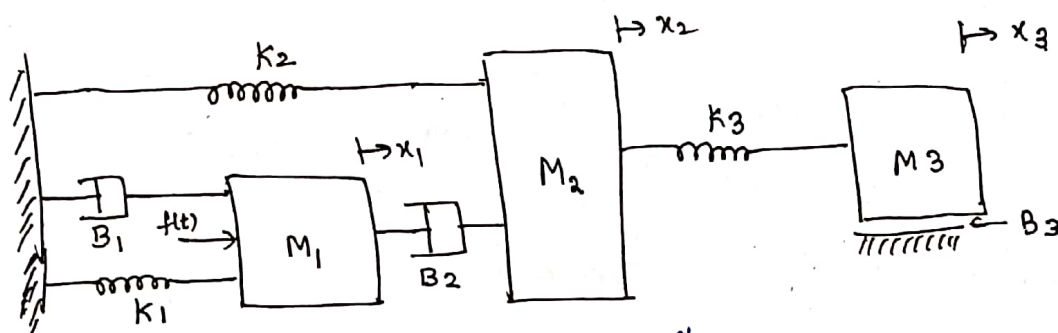
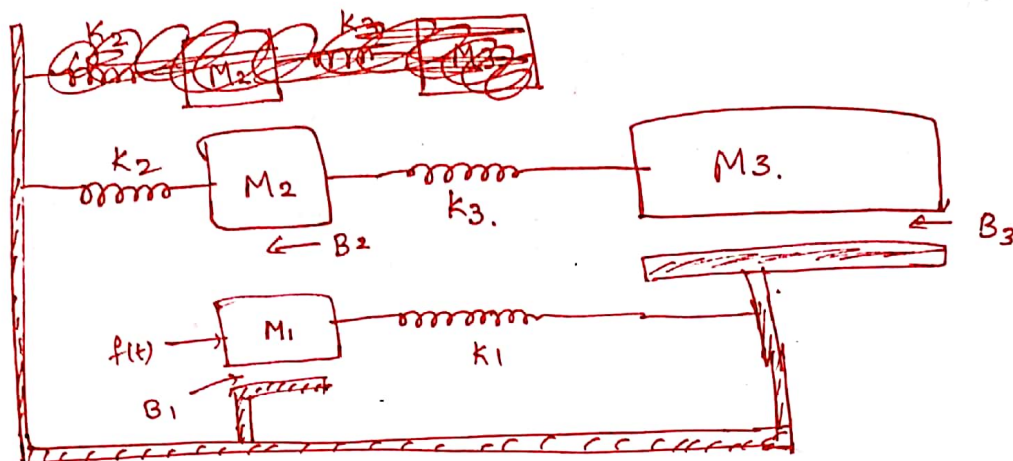


FV - analogy.



FI - Analogy

14



$$M_1 \ddot{x}_1 + B_1 \dot{x}_1 + k_1 x_1 + B_2 (\dot{x}_1 - \dot{x}_2) = f(t) \rightarrow (1)$$

$$L_1 \frac{di_1}{dt} + R_1 i_1 + \frac{1}{C_1} \int i_1 dt + R_2 (i_1 - i_2) = v(t)$$

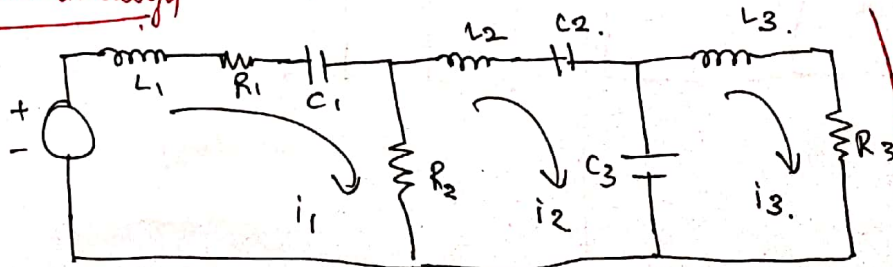
$$M_2 \ddot{x}_2 + k_2 x_2 + B_2 (\dot{x}_2 - \dot{x}_1) + k_3 (x_2 - x_3) = 0 \rightarrow (2)$$

$$L_2 \frac{di_2}{dt} + \frac{1}{C_2} \int i_2 dt + R_2 (i_2 - i_1) + \frac{1}{C_3} \int (i_2 - i_3) dt = 0$$

$$M_3 \ddot{x}_3 + B_3 \dot{x}_3 + k_3 (x_3 - x_2) = 0 \rightarrow (3)$$

$$L_3 \frac{di_3}{dt} + R_3 i_3 + \frac{1}{C_3} \int (i_3 - i_2) dt = 0$$

FV-analogy



FI analogy

